

Student Answer Script View



058-END SEMSTER-ODD-2025

Answer Sheet

Course: Computer Science and Engineering**Year/Sem:** Semester 7**Subject Name:** Introduction To Extended Reality Application Development and Digital Twins**28.00****50.00****Exam Date:** 16-Nov-2025

Comments submitted by student.

Q.No : 1 A)

Score : 2.00 / 3.00

Explain the concept of Extended Reality and summarize how it integrates with Digital Twin technology.

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Answer: i) Extended Reality refers to the combination of Virtual Reality + Mixed Reality + Augmented Reality together. $XR = VR + AR + MR$.

ii) It is the way where users can completely immerse themselves in a virtual environment for various applications in healthcare, entertainment, Education and Training.

iii) Extended Reality integrates with digital twins by overlaying information in the real world thus making it a revolutionary combination for Real world application.

Q.No : 1 B)

Score : 3.00 / 3.00

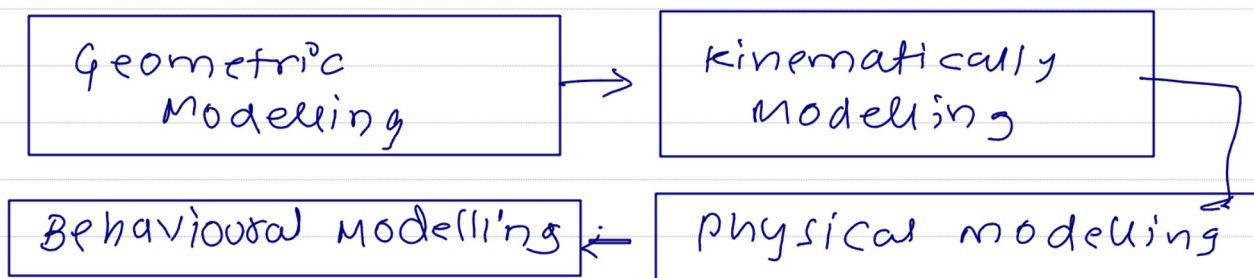
Differentiate among the four types of modelling used in XR and illustrate their relevance in rendering.

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Answer:- The Four types of modelling that is used in XR are Geometric Modelling, kinematical modelling, physical modelling / dynamic modelling and Behavioural modelling

1. Geometric Modelling:- It is used to define, the shape, size and structure of the object
uses:- It is used in rendering for determining the shape size and structure of the robotic arm.
2. kinematical Modelling:- used to define the motion of the rigid body (without the forces)
uses:- It is used in rendering for determining a rigid body's motion without forces.
3. Dynamic Modelling/ physical modelling:- It is used to define the motion of the rigid body (with the forces)
uses:- rendering the rigid body with the forces such as physics and torque applied
4. Behavioural Modelling:- used to define the behaviour of the rigid body. or action
uses:- renders behaviour



Q.No : 1 C)

Score : 4.00 / 4.00

Scenario: You are part of a startup building a remote-training simulator using VR headsets. Analyze the essential hardware components required to achieve realistic immersion and justify your selection.

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Answer:- JFI was a startup building a remote-training simulator using VR headset here is all the hardware components to achieve realistic immersion:-

1. Displays (OLED display)
2. Lenses (purely pancake lenses).
3. Accelerometers, and Gyroscope
4. Haptic Feedback vibrators (for immersive experience)
5. Speakers : : (for Immersive 3D Audio)
6. Infra-red camera (for tracking Eye movements)

Justification:-

Displays:- I would specifically use a OLED display (organic Light Emmiting diode) display as it would give us high contrast, great colours which would completely immerse the training experience, as the graphics would appear vivid with better viewing angles (Field of view).

Lenses : we can choose any lenses like Fresnel or pancake lenses, but I specifically choose pancake lenses because, it completely avoids us from ghosting and God rays, as its compact and optically folded, which will make the VR headset high quality.

Accelerometers & Gyroscope :- These motion tracking sensors give us the opportunity for better

motion and position tracking of the simulation of training, thus we can simulate the real world, for more realistic and Enhanced immersive training

Speakers: Speakers play a crucial part in our VR Headset, as they will help us in motion tracking and movement in the Virtual Environment giving a more realistic and 3D audio imitation of the real world.

Infra-red camera: This sensor is also crucial because, its going to track the movements of my eyes for a better understanding of the movements and actions immitating real life, best sensor that can be used in VR headset for remote-training

Q.No : 2 A)

Score : 3.50 / 4.00

Explain the concept of embodiment and discuss how it affects user presence in the Metaverse.

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Answer:- Embodiment: Fully immersed VR experience in the virtual world imitating real life body is known as embodiment.

It affects the user presence in metavers via:-

1. Self ownership:- The user starts to immitate its real body in the virtual world thus acheiving self ownership.
2. Agency:- This allows the user to do certain actions or move and navigate in the virtual world thus achieving agency in virtual environment.
3. Body-Interaction: Allow the user to immersify with the virtual body as its body movements

Q.No : 3 A)

Score : 2.00 / 3.00

Explain multimodal interaction in XR and identify advantages of gesture-based interfaces.

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Answer:- Multimodal Interaction In XR:-

multimodal

- i) Interactions in XR can be defined as a set of Actions, which can control the switching of interfaces with the help of movements of user.
- ii) One of the key advantage of gesture based interface, is that, we are not required to physically touch a virtual screen to change the interfaces.
- iii) An infra-red camera can track the pointing of your eyeball to change the interface of an application.

Q.No : 3 B)

Score : 3.00 / 3.00

Discuss how graphics, sound, and haptic feedback displays enhance user immersion.

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Solution:-

1. Graphics :- A good graphics in a game play simulates a more close imitation of the real world, thus enhancing the overall experience of a certain environment. It also helps in helping the user to be completely immersed in the virtual environment, making it more realistic and enjoyable.

2. Sound (audio).

3D audio: which helps in tracking the position, location and the distance of a certain body thus helping us achieve a complete imitation of the real world audio, which is in 3D. This helps users enhance their experience imitating real life.

3. Feedback displays: Feedback displays containing vibration motors helps in giving trigger vibration, when shooting, reloading a gun, which helps users to completely enhance their gaming or cinematic experience in the virtual environment.

Q.No : 3 C)

Score : 0.00 / 4.00

Scenario: In a VR rehabilitation system, users interact with virtual objects using hand gestures. Analyze how navigation and manipulation interfaces can be designed to maximize comfort and minimize fatigue.

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Q.No : 4 A)

Score : 3.00 / 4.00

Describe multiple-camera tracking and discuss its application in full-body motion capture.

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Answer:- Multiple-camera tracking is a process, where multiple AR cameras are setup in a scene, to track and deploy users for Full-body motion capture:-

i) When an AR/VR game is in production, game developers intend to put multiple AR cameras while designing a scene in a gaming engine

ii) It is done so, because, the gamers (who is going to play that game) can completely track its body movements and actions with complete physics for a better viewpoint angle to make the gaming experience much more immersive, realistic and enjoyable

iii) For example in game of "Call of duty", there are multiple AR cameras which can track the angles of body movements enabling the gamer to feel the action of the movement through the game.

iv) thus multiple-camera tracking helps in the application of full-body motion capture,

Q.No : 4 C)

Score : 1.50 / 3.00

Scenario: In an outdoor AR exploration app, evaluate how GPS-based tracking compares with SLAM for position accuracy.

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Answer: In an outdoor AR exploration app, GPS-based tracking and SLAM can effect the position Accuracy

GPS Tracking	SLAM (Simultaneous Location Access Monitoring)
Advantages:- i) Low computation power required ii) Efficient tracking iii) Accurate to 10 meters iv) Low complexity Disadvantages i) High Latency ii) Not Accurate to $< 1m$	Advantages:- i) Low-latency ii) Higher accuracy in terms of exact metrics of location iii) Accurate to $< 1m$ Disadvantages i) High computational power ii) Highest complexity

Q.No : 5 A)

Score : 3.00 / 3.00

Explain the role of a game engine in designing virtual worlds and summarize its key components.

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Answer:- Game Engine is Important in designing virtual world, it provides AR SDKs (standard development kits) and VR components that help design scenes and thus creating beautiful graphics and Environment which can immerse real world experiences in virtual Environment

Key components of a Game engine:-

1. XR plugins
2. AR SDKs (Vuforia/ARKit/ARCore)
3. VR components.

1. **XR plugins**: A XR plugin is one of the three important components in game engine. It provides important game components (example Gameobjects) to help create a VR/AR Games

2. **AR SDKs**: AR SDKs (Vuforia, ARKit/ARCore) are another component in a game engine, which provides the necessary support to build responsive AR Application in the field of motion tracking, light estimation and face detection.

3. **VR components.**: VR components help to contribute in a game engine for the necessary VR support for building VR applications, which provides Gameobjects navigation etc.

Q.No : 5 C)

Score : 3.00 / 4.00

Scenario: You are leading a team developing a mixed-reality educational game. Design a brief approach outlining how physics simulation and animation contribute to an engaging learning experience.

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Answer:- Approach on physics simulation and animation that can contribute to engaging learning experience:

1. Step 1:- First I would use the concept of kinematics and physics to show to interact with the team for designing a educational game.
2. Step 2:- To show the Lets say 'solar system' I would use rotation, scale of GameObjects of planets in an mixed reality environment for immersive experience of our solar system. in the virtual world.
3. Step 3:- then I will use the animate() function and add a delay to show the rotation and revolution of planets (physics) of it, thus showing each planets time and speed of revolution
4. Step 4:- I would add prefabs of cracks and other UI elements like rain and ice to make it more realistic
- Step 5:- then I would design the scenes add AR cameras for visualization, and complete the build.



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